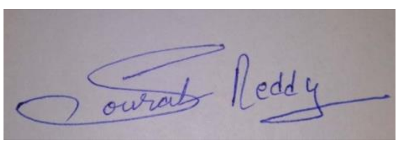
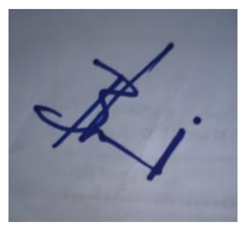
Project Summary

Batch details	PGPDSE-FT MUMBAI JUL22
Team members	Sakshi korde, Bhumi Baraiya, Sushane joshi, Kshitij salain, Rishi rajshekaran, Vishal kanojiya
Domain of Project	Finance & Risk Analytics
Proposed project title	Analysis of Loan Defaults.
Group Number	3
Team Leader	Bhumi Baraiya
Mentor Name	Mr. Sourab Reddy

Date: 26 November 2022





Signature of the Mentor

Signature of the Team Leader

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Project Details

Project is based on predicting the defaulting loans. The predictions are done by evaluating various factors that influence the repayment capacity of the loan applicant. The project aims to create a machine learning model which is highly accurate in predicting such defaults.

Business problem statement (GOALS)

1. Business Problem Understanding

Lending is one of the primary functions in banking operations. It plays a vital role in the growth of the business as well as economy. Lending always carries a huge risk along with it i.e., incapacity of repayments, defaults etc and thus it becomes important to analyse the repayment capacity of the applicant. Analysis of such defaults takes into account various factors such as income, loan tenure, credit scores etc. With the help of all the significant factors we predict the defaulting loans.

2. Business Objective

The objective is the assessment of credit risk from a chosen set of machine learning techniques, which technique exhibits the best performance in default prediction with regards to a specific model evaluation metric.

3. Approach

To Understand the data and observing all the relevant information,
To clean the data and perform the required EDA,
Further performing the necessary scaling and encoding if required,
Training the model using various Classification/Regression machine learning algorithms to predict and analyze the loan defaults.

4. Conclusions

Conclusion is to obtain a machine learning model which is highly accurate in predicting the loan defaults.

TOPIC SURVEY IN BRIEF

1. Problem understanding

Lending is very risky in that repayment of the loans is not always guaranteed and most of the times depend on other factors not in the control of the borrower. Therefore, managing loans in a proper way not only has positive effect on the bank's performance but also on the borrower and a country's economy as a whole. Failure to manage loans, which make up the largest share of banks assets, would likely lead to high levels of non -performing loans. And this in turn effect on the performance of Banks and the economy at large.

Regular monitoring of loan quality, possibly with an early warning system capable of alerting regulatory authorities of potential bank stress, is thus essential to ensure a sound financial system and prevent systemic crises

2. Current solution to the problem

Banks' requirement for collaterals when making loans reduces the adverse selection problem, which in turn leads to lower default rate and also evaluates the application using their individual evaluation method to determine the credit worthiness of the company.

3. Proposed solution to the problem

The defaults can be minimized by evaluating the repayment capacity of the person by considering various factors which leads to defaulting that are annual income, credit score, total active loans (primary as well as secondary), loan tenure, etc.
These predictions can be made with the help of machine learning models. The decision of granting loans can be taken with the help of such model if the model says yes then the loan can be granted.

4. Reference to the problem

Default is the failure to make required interest or principal repayments on a debt, whether that debt is a loan or a security. Individuals, businesses, and even countries can default on their debt obligations. [Default risk] is an important consideration for creditors.

CRITICAL ASSESSMENT OF TOPIC SURVEY

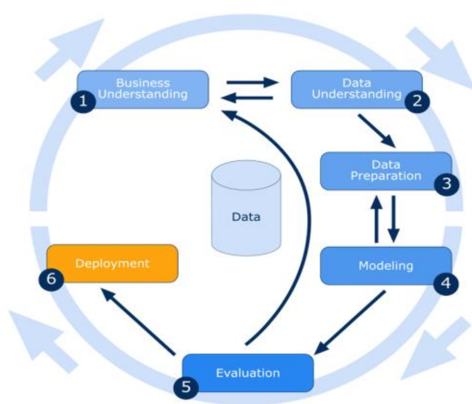
1. Find the key area, gaps identified in the topic survey where the project can add value to the customers and business

The key areas and gaps in the project are the determinant factors that affects the credit risk. The project can help in evaluating the credit risk of the applicant and also can provide the suggestion for the decision of granting of loans.

2. What key gaps are you trying to solve?

The gaps we are trying to solve are choosing wisely the factors, features and evaluating them in a manner that they helps us in giving the accurate predictions.

METHODOLOGY TO BE FOLLOWED



1. **Business understanding**

We would head start our project with understanding the business and getting to know about the domain. Will go through the articles and all the related information which will help us in better decision making, evaluation and analysis.

2. Data Understanding

After grabbing all the domain knowledge, we will lead towards the understanding of the data. Will then prepare the data dictionaries which will describe every feature in the dataset. The data understanding will help us in understanding our scope of project and thus help in decision making.

3. Data Preparation

Once we completely understand the dataset provided, further we will start its preparation for the model. We will clean it. Checking all the datatypes making sure they are in their correct form. Then we will check the distribution of the dataset and also the missing values, fill or drop those values according to the need. Further we will check for the outliers/extreme observations influencing the data. After processing the outliers, we will thus scale or encode if necessary for the model preparation.

4. Modelling

For modelling we will consider the processed data and split it into train and test data. The train data is used to train the model and the test data is used for checking the accuracy of the model. After splitting we will consider the train data and fit it into various classification models. Once the model is built, we will fit the model in test data and predict the outcome. After receiving the outcome, we will compare the predicted values and the actual values. The difference in the values will help us and calculating the accuracy of the model.

5. Evaluation

After the model building is done, we will compare each and every model with every other model created. We will compare the accuracy of the models which will help us in determining the model with the most correct predictions. Further if the model needs changes, we will tune the parameters to gain the highest accuracy. The model with the least errors will thus be finalized.

REFERENCES

https://www.projectclue.com/banking-and-finance/project-topics-materials-for-undergraduate-students/an-assessment-of-loan-defaults-and-its-impacts-on-profitability-of-bank

https://medium.com/@arbasith/dsi-capstone-project-predicting-loan-defaults-4ebbad3292b4

Notes For Project Team

Original owner of data	AVIK PAUL
Data set information	Vehicle Loan Default Prediction
Any past relevant articles using the dataset	Unknown
Reference	Kaggle
Link to web page	https://www.kaggle.com/datasets/avikpaul4u/vehicle-loan-default-prediction?select=Data+Dictionary.xlsx

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